

What is Big Data?

Sure! Let's talk about Big Data in simple terms.

Big Data refers to the enormous amounts of information that we generate every day in our digital world. Imagine a giant library filled with millions of books, but instead of just books, it contains videos, images, social media posts, and more. This data is collected continuously and comes in many different forms, making it complex and challenging to manage. Unlike Small Data, which is easy to understand and can be stored in a simple file, Big Data requires special tools and techniques to analyze and make sense of it.

To help us understand Big Data better, we can think of it in terms of the "five V's": Volume (the sheer amount of data), Velocity (how fast the data is generated), Variety (the different types of data), Veracity (the accuracy of the data), and Value (the insights we can gain from it).

What is Big Data?

"The basic idea behind the phrase 'Big Data' is that everything we do is increasingly leaving a digital trace (or data), which we can use and analyze to become smarter. The driving forces in this brave new world are access to ever-increasing volumes of data and our ever-increasing technological capability to mine that data for commercial insights."

—Bernard Marr

Big Data vs. Small Data

Small Data

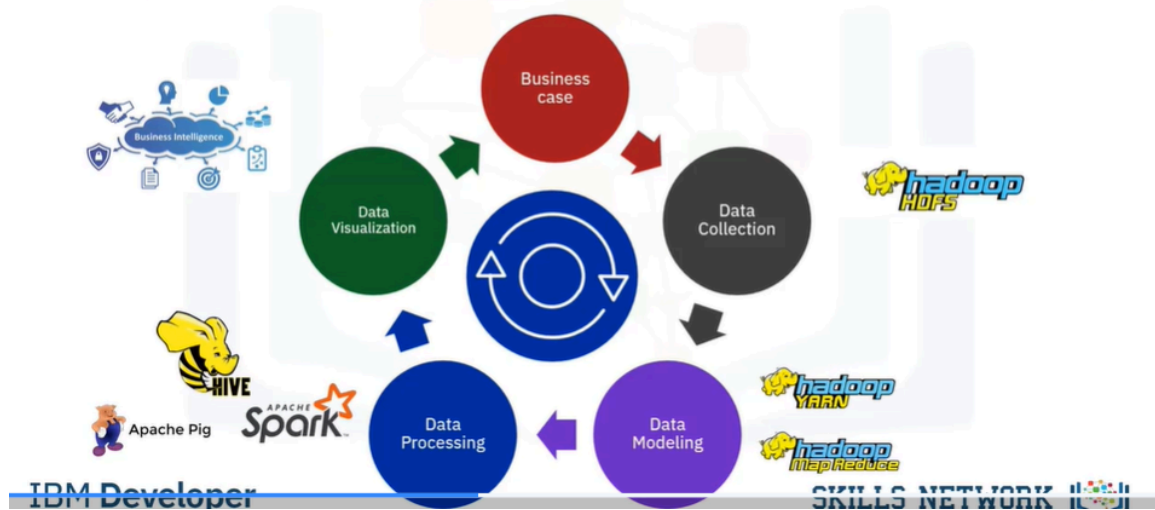
- Small enough for human inference
- Accumulated slowly
- Relatively consistent and structured data usually stored in known forms such as JSON and XML
- Mostly located in storage systems within Enterprises or data centers

Big Data

- Data generated in huge volumes and could be structured, semi-structured, or unstructured
- Needs processing to generate insights for human consumption
- Arrives continuously at enormous speed from multiple sources
- Comprises any form of data including video, photos, and more
- Distributed on the cloud and server farms

- To understand Big Data, we need to compare it with Small Data. Small Data is available in limited quantities that can be easily interpreted by humans with little or no digital processing. Sports scores and employee shift schedules are some examples of Small Data. It is accumulated slowly and may or may not get updated continuously. Storing Small Data is relatively easy as it is mostly available in a structured format such as First and Last Name, Address, Gender, and so on, and It can be maintained within an enterprise's own infrastructure or in a data center.
- Big Data, on the other hand, is generated in massive volumes and has little or no structure in which it is received. Examples of semi-structured data include social media posts that could be images accompanied by hashtags, while unstructured data could include medical records from millions of patients. It is complex and requires specialized programs to interpret and make it available for human consumption. Not only is Big Data massive in size, it is collected continuously and grows exponentially in a short amount of time. Big Data could be in any form including but not limited to text, images, audio, and videos. Finally, Big Data is so voluminous that it is stored in the Cloud or on server farms set up specifically for this purpose.

Big Data life cycle



- It is a common misconception that Big Data refers to just large volumes of data. In reality, Big Data is the entire life cycle of working with large volumes of data. Let's take a look at each phase in the Big Data life cycle. Big Data collection is initiated as a result of a business problem or requirement. As data is collected, it gets stored using a framework for distributed storage such as Hadoop HDFS. To make sense of all the data collected, Map and Reduce tasks and scripts create a data model to store it in a database.
- This data model includes the various data entities (or objects), and the relationship and rules between these entities. After modeling, data is ready to be processed. Tools such as Apache Spark are used to produce meaningful information from the modeled data. Finally, the processed data is visualized and presented in a graphical format such as charts and graphs. This visualized data is then used for making meaningful business decisions and lead to new business cases, thereby creating a continuous life cycle.

Characteristics of Big Data

"Big Data is high-**volume**, high-**velocity** and/or high-**variety** information assets that demand cost-effective, innovative forms of information processing that enable enhanced insight, decision making, and process automation."

—Gartner

- The research firm Gartner defines Big Data as a high-volume, high-velocity, and high-variety information asset that demands cost-effective and innovative tools for processing.

Bit, bytes and more

Bits = 0 or 1

8 Bits = 1 Byte

1024 Bytes = 1 Kilobyte(KB)

1024 KB = 1 Megabyte(MB)

1024 MB = 1 Gigabyte(GB)

1024 GB = 1 Terabyte(TB)

1024 TB = 1 Petabyte(PB)

1024 PB = 1 Exabyte(EB)

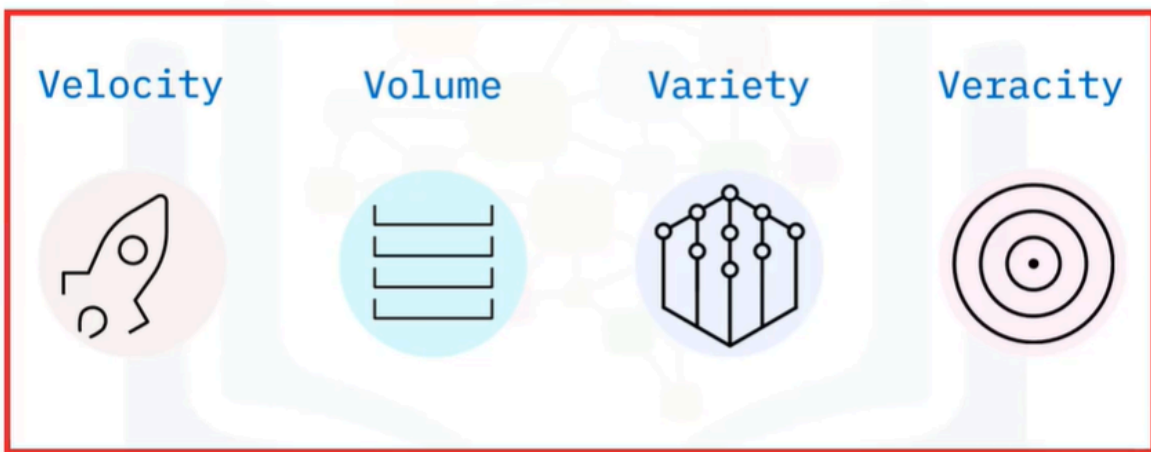
1024 EB = 1 Zettabyte(ZB)

1024 ZB = 1 Yottabyte(YB)

- Let's take a moment to appreciate just how huge Big Data volumes really are. You have probably heard of megabytes, gigabytes, terabytes, and possibly

even petabytes. But big data can be even bigger. Do you know you can store over 11 million movies in 4k resolution in just one exabyte of space? Now visualize this... modern computers generally ship with 1- to 5-terabyte hard drives. One zettabyte contains a billion terabytes, while a yottabyte fits in one trillion terabytes?

The four V's of Big Data



- Doesn't that boggle the mind? When we talk about Big Data, we traditionally talk about the four V's of Big Data. These are: Velocity, Volume, Variety, and Veracity. Velocity is the speed at which data arrives. Volume is the increase in the amount of data stored over time.

Variety



Description

- Data that comes from machines, people, and processes
- Structured, semi-structured, and unstructured data

Attributes

- Structure, complexity, and origin

Drivers

- Mobile technologies
- Scalable infrastructure
- Resilience
- Fault recovery
- Efficient storage and retrieval

- Variety is the diversity of data. Many forms of data exist and need to be stored. Veracity is the certainty of data. With a large amount of data available, how will we know if the data collected is accurate or inaccurate? These four main components are used to describe the dimensions of Big Data.

Velocity



Description

- Data that is generated fast
- Process that never stops

Attributes

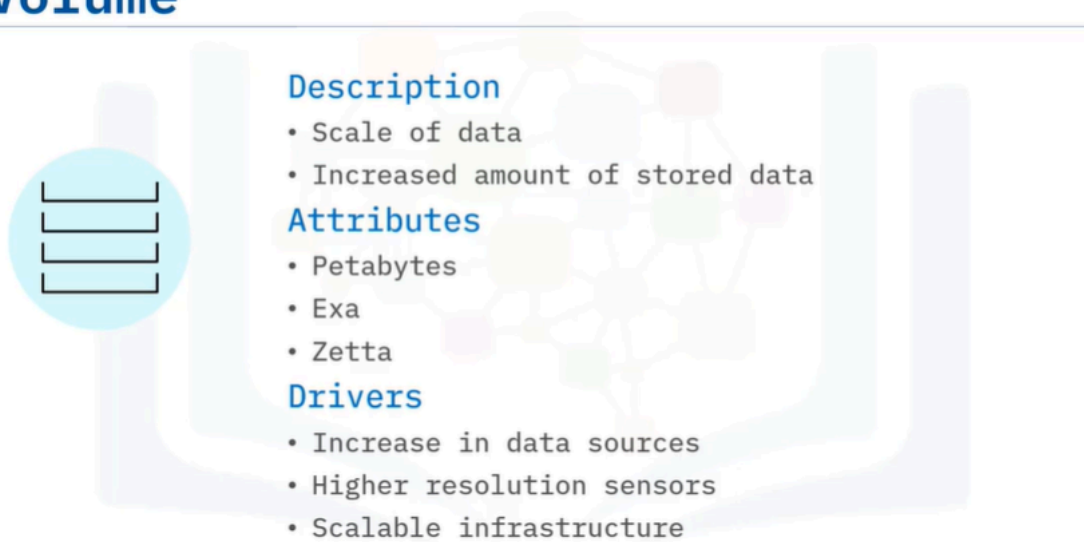
- Batch
- Close to real time
- Streaming

Drivers

- Improved connectivity and hardware
- Rapid response times
- Precalculated analysis

- Velocity signifies that data is being generated extremely fast, and the process never stops. Data must be processed quickly so that decisions can be made at the speed with which the data arrives. Velocity's main attributes are: batch close to real-time, and streaming. What are the drivers? Definitely improving connectivity and hardware. Just think about all the devices that are connected through the Internet today and all the super-fast response times. Big Data also supports upscaling of pre-calculated analysis.

Volume



- Volume is the increase in the amount of data stored. The amount of Big Data generated is vast compared to traditional data sources. Volume attributes of Big Data are: Petabytes, Exa, and Zetta, to name just a few. Typical drivers of volume in Big Data are: the increase in data sources, higher resolution sensors, and scalable hardware infrastructure. Variety is the diversity of the data. Data is generated by people and processes through the use of machines, from both inside and outside an organization. Some of the data is structured and semi-structured, but most is unstructured. The main attributes are structure, complexity, and origin. Drivers of Variety in Big Data can be: mobile technologies, scalable hardware infrastructure, resilience, fault recovery, and efficient storage and retrieval.

Veracity



Description

- Quality, origin, and conformity of facts
- Accuracy of data
- Data that comes from people and processes

Attributes

- Consistency and completeness
- Integrity
- Ambiguity

Drivers

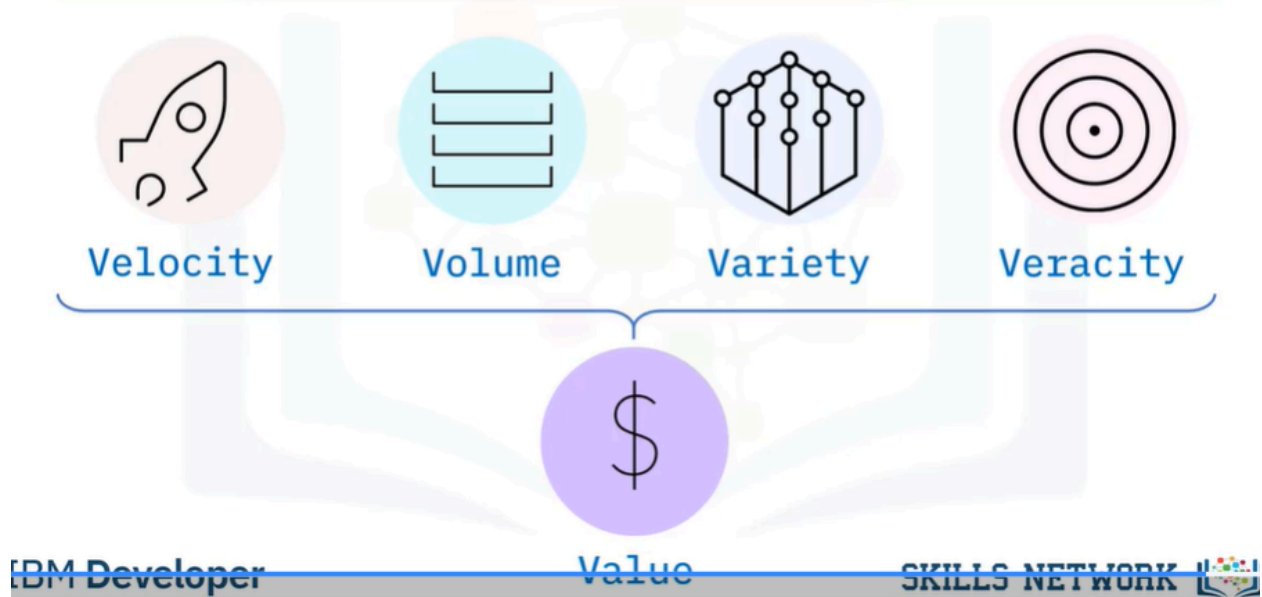
- Cost and traceability
- Robust ingestion
- ETL mechanisms

IM Developer

SKILLS NETWORK

- Veracity is the quality, origin, and conformity to facts and accuracy of the data. This is because data comes from both within and outside an organization. Attributes include consistency and completeness, integrity, and ambiguity. Drivers of Veracity in Big Data are: cost and the need for traceability, robust ingestion, and extract, transform, load (ETL) mechanisms. Big Data has another V that must be considered.

Adding the fifth V of Big Data



- The fifth V of Big Data is Value. It is the outcome of making intelligent business decisions from leveraging the previous four V's. The ultimate goal of an organization is to: produce value in the form of faster and smarter business decisions, increase efficient use of resources, and discover of new market opportunities. Big Data supports innovation and thus creates value.

Summary

In this video, you learned that:

- Big Data is the digital trace that gets generated through the entire digital ecosystem
- Big Data is a high-volume, high-velocity, and high-variety information asset
- The core features of Big Data are the four V's
- Big Data creates a fifth V: Value